

Lab 1. Network Scanning

1: Introduction

In this lab, we will perform network scanning on Kali, a Linux distribution designed for network security and penetration testing. Turn on all your VMs and log into your Kali Linux VM; it will be named MCY 650-001_username_Kali. Your Metasploitable VM will be used as our target; it will be named MCY 650-001_username_Meta. Its IP address will be designated by TARGET_IP below.

References

1. nmap documentation, <http://nmap.org/docs.html>
2. Secrets of Network Cartography

2: Ping Scans

We will run nmap in the terminal window. Find your Kali VM's IP address before proceeding. Your IP address will be used in this and later sections of the lab.

2.1: (9 points) Use a ping scan to check which of the 256 IP addresses in a /24 address space around your IP address are up. How many IP addresses are up in this range? Don't count manually. Pipe the output of nmap to a command that will count for you. **Include a screenshot of the commands you used, including the piped one used to get the host count.**

```
# nmap -sn YOUR_IP_ADDRESS/24
```

```
root@nku-kali:~# ifconfig
```

```
root@nku-kali:~# nmap -sn 10.2.60.3/24
```

```
root@nku-kali:~# nmap -sn 10.2.60.3/24 | grep -c "Host is up"
121
```

2.2: (9 points) List the highest and lowest IP addresses that were shown to be up by the ping scan. The awk command is useful for parsing nmap output, as it can find the lines that you want using a pattern in //s and then select the column of output that contains the information you want to retrieve. Include screenshots of the top and bottom part of your output.

```
# nmap -sn YOUR_IP_ADDRESS/24 | awk '/Nmap scan report/ {print $5}'
```

```
root@nku-kali:~# nmap -sn 10.2.60.3/24 | awk '/Nmap scan report/ {print $5}'
```

Highest: 10.2.60.250

Lowest: 10.2.60.49

3: Port Scans

In this section, we will scan TCP ports in several different ways to gather information on an example target system. We will scan UDP ports at the end of the lab, as these scans take a long time to complete.

3.1: (9 points) What ports are open on the scanme.nmap.org test server? Use a TCP connect scan. Your answer must include only the port numbers. **Include a screenshot of them.**

```
# nmap -sT scanme.nmap.org
```

PORT	STATE	SERVICE
22/tcp	open	ssh
25/tcp	filtered	smtp
80/tcp	open	http
445/tcp	filtered	microsoft-ds
465/tcp	filtered	smtps
6881/tcp	filtered	bittorrent-tracker
9929/tcp	open	nping-echo
31337/tcp	open	Elite

3.2: (9 points) Using a TCP SYN scan, what ports do you find open on scanme.nmap.org? **Include a screenshot.**

```
# nmap -sS scanme.nmap.org
```

PORT	STATE	SERVICE
22/tcp	open	ssh
25/tcp	filtered	smtp
80/tcp	open	http
445/tcp	filtered	microsoft-ds
465/tcp	filtered	smtps
6881/tcp	filtered	bittorrent-tracker
9929/tcp	open	nping-echo
31337/tcp	open	Elite

3.3: (9 points) Looking at the output of the two scans outside the ports listed, what differences do you find between the TCP connect and SYN scans? If there is no difference, then just write “None” below.

None

3.4: (9 points) Some machines are behind a firewall, which filters connections to some ports, preventing nmap from receiving any response from those ports. Blocked ports may be listed as either “filtered” or “closed”. To see an example of such a scan, perform a TCP connect scan on www.example.com. Your answer must include any listed port numbers for closed and open ports. **Include a screenshot.**

```
# nmap -sT www.example.com
```

PORT	STATE	SERVICE
80/tcp	open	http
443/tcp	open	https
1119/tcp	closed	bnetgame
1935/tcp	closed	rtmp

3.5: (9 points) To determine why a scan returns the results that it does, use the --reason option. Explain the reasons that ports are listed as open, closed, or filtered in the scan of www.example.com.

```
# nmap --reason -sT www.example.com
```

PORT	STATE	SERVICE	REASON
80/tcp	open	http	syn-ack
443/tcp	open	https	syn-ack
1119/tcp	closed	bnetgame	conn-refused
1935/tcp	closed	rtmp	conn-refused

3.6: (9 points) To see every packet sent by a scan, use the `--packet-trace` option. We will save this output in a file for further analysis using I/O redirection. We do not redirect `STDERR`, so we will still see error output on the screen. In the box below, explain what packet is sent first in the scan and count how many packets are sent in total. Use a command to do the counting for you, but be sure not to count regular nmap output (what you would see without the trace option). **Include your command in your answer.**

```
# nmap --packet-trace -sS scanme.nmap.org > trace.out
# less trace.out
```

The first packet sent is a ICMP Echo request Type 8/ Code 0 (which is standard w/ no subtypes or variations) from my Kali Linux VM IP address to the scanme.nmap.org server IP address.

```
root@kali:~# grep -i "TCP" trace.out | wc -l
2011
```

3.7: (10 points) The nmap scanner can return additional information, including service versions, OS identification, and tracerouting. The `-A` option will perform all of these tests. The `-T4` option tells nmap to use aggressive packet timing, which can be dangerous as it can cause some older machines to crash. However, scanme.nmap.org is configured so that it will have no problems with the `-T4` option. Even with the faster speed, this scan will take longer than previous ones due to the large number of tests performed. We add the `-v` option so that you can see the scan in progress

```
# nmap -v -A -T4 scanme.nmap.org | less
```

Based on the output of the scan above, answer the following questions:

1. What is the server software name and version for each of the ports?

```
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
| 1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
| 2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
| 256 96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_ 256 33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open  http         Apache/2.4.7 (Ubuntu)
|_ http-server-header: Apache/2.4.7 (Ubuntu)
445/tcp    filtered microsoft-ds
465/tcp    filtered smtps
6881/tcp   filtered bittorrent-tracker
9929/tcp   open  nping-echo    Nping echo
31337/tcp  open  tcpwrapped
```

2. What title would you see in the top of your web browser if you contacted the web server at scanme.nmap.org?

https://nmap.org

3. How many network hops does it take to reach scanme.nmap.org from your VM?

Network Distance: 4 hops

3.8: (9 points) By default, nmap scans the most common 1000 ports. With the fast option, nmap scans only 100 ports. Using the -p option, we can configure nmap to configure specific sets of ports, ranging from 1 port to the entire 65,536 possible ports. Scanning all ports can take around 10 minutes over the Internet, so we will scan a local server for this question. How many open ports did you find for each of the scans? How many seconds did it take for each port scan to finish?

time nmap -F -sS TARGET_IP

```
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

time nmap -sS TARGET_IP

```
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds
```

time nmap -p0-65535 -sS TARGET_IP

```
PORT      STATE SERVICE
22/tcp    open  ssh
111/tcp    open  rpcbind
Nmap done: 1 IP address (1 host up) scanned in 0.92 seconds
```

3.9: (9 points) UDP scans are much slower than TCP scans due to the unreliability of UDP, so the scan in this question will require the longest amount of time. You may want to work on another assignment while this scan is running. Write the list of open UDP ports in the box below. **Include all the output of the command.**

nmap -v -sU scanme.nmap.org

Port 123/udp 'open' 'ntp'

```
Starting Nmap 7.80 ( https://nmap.org ) at 2024-09-07 00:31 EDT
Initiating Ping Scan at 00:31
Scanning scanme.nmap.org (45.33.32.156) [4 ports]
Completed Ping Scan at 00:31, 0.04s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 00:31
Completed Parallel DNS resolution of 1 host. at 00:31, 0.00s elapsed
Initiating UDP Scan at 00:31
Scanning scanme.nmap.org (45.33.32.156) [1000 ports]
Discovered open port 123/udp on 45.33.32.156
Completed UDP Scan at 00:32, 4.05s elapsed (1000 total ports)
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0062s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 999 open|filtered ports
PORT      STATE SERVICE
123/udp    open  ntp

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 4.21 seconds
Raw packets sent: 2006 (58.050KB) | Rcvd: 5 (236B)
```